

4. Root Cause Analysis (RCA) Archive

```
=====
Incident Report: INC-402-0
Subject:         Payment Gateway Timeout
Date:           2024-01-16
Duration:       98 Minutes
Severity:       Sev-1
=====
```

1. Problem Statement

The monitoring system alerted high latency and error rates in the Payment Gateway Timeout module.
Users experienced 503 errors.

2. Root Cause Analysis (5 Whys)

- Why failed? The connection pool was exhausted.
- Why exhausted? Threads were waiting on I/O.
- Why waiting? The external vendor API was slow.
- Why slow? Vendor was undergoing maintenance.
- Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

4. Prevention / Action Items

- [] Implement Hystrix Circuit Breaker (Due: Jan 30)
- [] Increase timeout thresholds (Due: Jan 25)
- [] Add alerting for connection pool saturation.

5. Timeline

08:00 - Alert fired
08:05 - Sev-1 declared
08:15 - Root cause identified
08:30 - Fix deployed

```
=====
Incident Report: INC-404-1
Subject:         Memory Exhaustion in Report Pod
Date:           2024-01-15
Duration:       33 Minutes
Severity:       Sev-1
=====
```

CONFIDENTIAL - INTERNAL USE ONLY

1. Problem Statement

The monitoring system alerted high latency and error rates in the Memory Exhaustion in Report Pod module.

Users experienced 503 errors.

2. Root Cause Analysis (5 Whys)

- Why failed? The connection pool was exhausted.
- Why exhausted? Threads were waiting on I/O.
- Why waiting? The external vendor API was slow.
- Why slow? Vendor was undergoing maintenance.
- Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

4. Prevention / Action Items

- [] Implement Hystrix Circuit Breaker (Due: Jan 30)
- [] Increase timeout thresholds (Due: Jan 25)
- [] Add alerting for connection pool saturation.

5. Timeline

08:00 - Alert fired

08:05 - Sev-1 declared

08:15 - Root cause identified

08:30 - Fix deployed

=====

=====

Incident Report: INC-403-2

Subject: DDoS Attack on API

Date: 2024-01-19

Duration: 49 Minutes

Severity: Sev-1

=====

1. Problem Statement

The monitoring system alerted high latency and error rates in the DDoS Attack on API module.

Users experienced 503 errors.

2. Root Cause Analysis (5 Whys)

- Why failed? The connection pool was exhausted.
- Why exhausted? Threads were waiting on I/O.
- Why waiting? The external vendor API was slow.
- Why slow? Vendor was undergoing maintenance.
- Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

CONFIDENTIAL - INTERNAL USE ONLY

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

4. Prevention / Action Items

- [] Implement Hystrix Circuit Breaker (Due: Jan 30)
- [] Increase timeout thresholds (Due: Jan 25)
- [] Add alerting for connection pool saturation.

5. Timeline

08:00 - Alert fired
08:05 - Sev-1 declared
08:15 - Root cause identified
08:30 - Fix deployed

=====

=====

Incident Report: INC-402-3
Subject: Payment Gateway Timeout
Date: 2024-01-2
Duration: 98 Minutes
Severity: Sev-1

=====

1. Problem Statement

The monitoring system alerted high latency and error rates in the Payment Gateway Timeout module.

Users experienced 503 errors.

2. Root Cause Analysis (5 Whys)

- Why failed? The connection pool was exhausted.
- Why exhausted? Threads were waiting on I/O.
- Why waiting? The external vendor API was slow.
- Why slow? Vendor was undergoing maintenance.
- Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

4. Prevention / Action Items

- [] Implement Hystrix Circuit Breaker (Due: Jan 30)
- [] Increase timeout thresholds (Due: Jan 25)
- [] Add alerting for connection pool saturation.

5. Timeline

08:00 - Alert fired
08:05 - Sev-1 declared

CONFIDENTIAL - INTERNAL USE ONLY

08:15 - Root cause identified

08:30 - Fix deployed

=====

=====

Incident Report: INC-402-4

Subject: Payment Gateway Timeout

Date: 2024-01-10

Duration: 75 Minutes

Severity: Sev-1

=====

1. Problem Statement

The monitoring system alerted high latency and error rates in the Payment Gateway Timeout module.

Users experienced 503 errors.

2. Root Cause Analysis (5 Whys)

- Why failed? The connection pool was exhausted.

- Why exhausted? Threads were waiting on I/O.

- Why waiting? The external vendor API was slow.

- Why slow? Vendor was undergoing maintenance.

- Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

4. Prevention / Action Items

- [] Implement Hystrix Circuit Breaker (Due: Jan 30)

- [] Increase timeout thresholds (Due: Jan 25)

- [] Add alerting for connection pool saturation.

5. Timeline

08:00 - Alert fired

08:05 - Sev-1 declared

08:15 - Root cause identified

08:30 - Fix deployed

=====

=====

Incident Report: INC-403-5

Subject: DDoS Attack on API

Date: 2024-01-25

Duration: 61 Minutes

Severity: Sev-1

CONFIDENTIAL - INTERNAL USE ONLY

1. Problem Statement

The monitoring system alerted high latency and error rates in the DDoS Attack on API module.

Users experienced 503 errors.

2. Root Cause Analysis (5 Whys)

- Why failed? The connection pool was exhausted.
- Why exhausted? Threads were waiting on I/O.
- Why waiting? The external vendor API was slow.
- Why slow? Vendor was undergoing maintenance.
- Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

4. Prevention / Action Items

- ☐ Implement Hystrix Circuit Breaker (Due: Jan 30)
- ☐ Increase timeout thresholds (Due: Jan 25)
- ☐ Add alerting for connection pool saturation.

5. Timeline

08:00 - Alert fired
08:05 - Sev-1 declared
08:15 - Root cause identified
08:30 - Fix deployed

Incident Report: INC-402-6

Subject: Payment Gateway Timeout

Date: 2024-01-20

Duration: 118 Minutes

Severity: Sev-1

1. Problem Statement

The monitoring system alerted high latency and error rates in the Payment Gateway Timeout module.

Users experienced 503 errors.

2. Root Cause Analysis (5 Whys)

- Why failed? The connection pool was exhausted.
- Why exhausted? Threads were waiting on I/O.
- Why waiting? The external vendor API was slow.
- Why slow? Vendor was undergoing maintenance.

CONFIDENTIAL - INTERNAL USE ONLY

- Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

4. Prevention / Action Items

- [] Implement Hystrix Circuit Breaker (Due: Jan 30)
- [] Increase timeout thresholds (Due: Jan 25)
- [] Add alerting for connection pool saturation.

5. Timeline

08:00 - Alert fired

08:05 - Sev-1 declared

08:15 - Root cause identified

08:30 - Fix deployed

=====

=====

Incident Report: INC-401-7

Subject: Database CPU Spike

Date: 2024-01-6

Duration: 47 Minutes

Severity: Sev-1

=====

1. Problem Statement

The monitoring system alerted high latency and error rates in the Database CPU Spike module.

Users experienced 503 errors.

2. Root Cause Analysis (5 Whys)

- Why failed? The connection pool was exhausted.
- Why exhausted? Threads were waiting on I/O.
- Why waiting? The external vendor API was slow.
- Why slow? Vendor was undergoing maintenance.

- Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

4. Prevention / Action Items

- [] Implement Hystrix Circuit Breaker (Due: Jan 30)
- [] Increase timeout thresholds (Due: Jan 25)
- [] Add alerting for connection pool saturation.

5. Timeline

CONFIDENTIAL - INTERNAL USE ONLY

08:00 - Alert fired
08:05 - Sev-1 declared
08:15 - Root cause identified
08:30 - Fix deployed

Incident Report: INC-401-8
Subject: Database CPU Spike
Date: 2024-01-19
Duration: 38 Minutes
Severity: Sev-1

1. Problem Statement

The monitoring system alerted high latency and error rates in the Database CPU Spike module.

Users experienced 503 errors.

2. Root Cause Analysis (5 Whys)

- Why failed? The connection pool was exhausted.
- Why exhausted? Threads were waiting on I/O.
- Why waiting? The external vendor API was slow.
- Why slow? Vendor was undergoing maintenance.

- Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

4. Prevention / Action Items

- [] Implement Hystrix Circuit Breaker (Due: Jan 30)
- [] Increase timeout thresholds (Due: Jan 25)
- [] Add alerting for connection pool saturation.

5. Timeline

08:00 - Alert fired
08:05 - Sev-1 declared
08:15 - Root cause identified
08:30 - Fix deployed

Incident Report: INC-405-9
Subject: Redis Cache Fragmentation
Date: 2024-01-5

CONFIDENTIAL - INTERNAL USE ONLY

Duration: 81 Minutes
Severity: Sev-1

=====

1. Problem Statement

The monitoring system alerted high latency and error rates in the Redis Cache Fragmentation module.

Users experienced 503 errors.

2. Root Cause Analysis (5 Whys)

- Why failed? The connection pool was exhausted.
- Why exhausted? Threads were waiting on I/O.
- Why waiting? The external vendor API was slow.
- Why slow? Vendor was undergoing maintenance.
- Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

4. Prevention / Action Items

- [] Implement Hystrix Circuit Breaker (Due: Jan 30)
- [] Increase timeout thresholds (Due: Jan 25)
- [] Add alerting for connection pool saturation.

5. Timeline

08:00 - Alert fired
08:05 - Sev-1 declared
08:15 - Root cause identified
08:30 - Fix deployed

=====

=====

Incident Report: INC-405-10
Subject: Redis Cache Fragmentation
Date: 2024-01-26
Duration: 63 Minutes
Severity: Sev-1

=====

1. Problem Statement

The monitoring system alerted high latency and error rates in the Redis Cache Fragmentation module.

Users experienced 503 errors.

2. Root Cause Analysis (5 Whys)

- Why failed? The connection pool was exhausted.
- Why exhausted? Threads were waiting on I/O.

CONFIDENTIAL - INTERNAL USE ONLY

- Why waiting? The external vendor API was slow.
- Why slow? Vendor was undergoing maintenance.
- Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

4. Prevention / Action Items

- [] Implement Hystrix Circuit Breaker (Due: Jan 30)
- [] Increase timeout thresholds (Due: Jan 25)
- [] Add alerting for connection pool saturation.

5. Timeline

08:00 - Alert fired

08:05 - Sev-1 declared

08:15 - Root cause identified

08:30 - Fix deployed

=====

=====

Incident Report: INC-403-11

Subject: DDoS Attack on API

Date: 2024-01-19

Duration: 22 Minutes

Severity: Sev-1

=====

1. Problem Statement

The monitoring system alerted high latency and error rates in the DDoS Attack on API module.

Users experienced 503 errors.

2. Root Cause Analysis (5 Whys)

- Why failed? The connection pool was exhausted.
- Why exhausted? Threads were waiting on I/O.
- Why waiting? The external vendor API was slow.
- Why slow? Vendor was undergoing maintenance.
- Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

4. Prevention / Action Items

- [] Implement Hystrix Circuit Breaker (Due: Jan 30)
- [] Increase timeout thresholds (Due: Jan 25)
- [] Add alerting for connection pool saturation.

CONFIDENTIAL - INTERNAL USE ONLY

5. Timeline

08:00 - Alert fired
08:05 - Sev-1 declared
08:15 - Root cause identified
08:30 - Fix deployed

=====

=====

Incident Report: INC-403-12
Subject: DDoS Attack on API
Date: 2024-01-8
Duration: 117 Minutes
Severity: Sev-1

=====

1. Problem Statement

The monitoring system alerted high latency and error rates in the DDoS Attack on API module.

Users experienced 503 errors.

2. Root Cause Analysis (5 Whys)

- Why failed? The connection pool was exhausted.
- Why exhausted? Threads were waiting on I/O.
- Why waiting? The external vendor API was slow.
- Why slow? Vendor was undergoing maintenance.
 - Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

4. Prevention / Action Items

- [] Implement Hystrix Circuit Breaker (Due: Jan 30)
- [] Increase timeout thresholds (Due: Jan 25)
- [] Add alerting for connection pool saturation.

5. Timeline

08:00 - Alert fired
08:05 - Sev-1 declared
08:15 - Root cause identified
08:30 - Fix deployed

=====

=====

Incident Report: INC-402-13

CONFIDENTIAL - INTERNAL USE ONLY

Subject: Payment Gateway Timeout
Date: 2024-01-7
Duration: 37 Minutes
Severity: Sev-1

=====

1. Problem Statement

The monitoring system alerted high latency and error rates in the Payment Gateway Timeout module.

Users experienced 503 errors.

2. Root Cause Analysis (5 Whys)

- Why failed? The connection pool was exhausted.
- Why exhausted? Threads were waiting on I/O.
- Why waiting? The external vendor API was slow.
- Why slow? Vendor was undergoing maintenance.

- Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

4. Prevention / Action Items

- [] Implement Hystrix Circuit Breaker (Due: Jan 30)
- [] Increase timeout thresholds (Due: Jan 25)
- [] Add alerting for connection pool saturation.

5. Timeline

08:00 - Alert fired
08:05 - Sev-1 declared
08:15 - Root cause identified
08:30 - Fix deployed

=====

=====

Incident Report: INC-404-14
Subject: Memory Exhaustion in Report Pod
Date: 2024-01-27
Duration: 95 Minutes
Severity: Sev-1

=====

1. Problem Statement

The monitoring system alerted high latency and error rates in the Memory Exhaustion in Report Pod module.

Users experienced 503 errors.

2. Root Cause Analysis (5 Whys)

CONFIDENTIAL - INTERNAL USE ONLY

- Why failed? The connection pool was exhausted.
- Why exhausted? Threads were waiting on I/O.
- Why waiting? The external vendor API was slow.
- Why slow? Vendor was undergoing maintenance.
 - Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

4. Prevention / Action Items

- [] Implement Hystrix Circuit Breaker (Due: Jan 30)
- [] Increase timeout thresholds (Due: Jan 25)
- [] Add alerting for connection pool saturation.

5. Timeline

08:00 - Alert fired
08:05 - Sev-1 declared
08:15 - Root cause identified
08:30 - Fix deployed

=====

=====

Incident Report: INC-404-15
Subject: Memory Exhaustion in Report Pod
Date: 2024-01-20
Duration: 20 Minutes
Severity: Sev-1

=====

1. Problem Statement

The monitoring system alerted high latency and error rates in the Memory Exhaustion in Report Pod module.

Users experienced 503 errors.

2. Root Cause Analysis (5 Whys)

- Why failed? The connection pool was exhausted.
- Why exhausted? Threads were waiting on I/O.
- Why waiting? The external vendor API was slow.
- Why slow? Vendor was undergoing maintenance.
 - Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

4. Prevention / Action Items

- [] Implement Hystrix Circuit Breaker (Due: Jan 30)

CONFIDENTIAL - INTERNAL USE ONLY

- [] Increase timeout thresholds (Due: Jan 25)
- [] Add alerting for connection pool saturation.

5. Timeline

08:00 - Alert fired
08:05 - Sev-1 declared
08:15 - Root cause identified
08:30 - Fix deployed

=====

=====

Incident Report: INC-402-16
Subject: Payment Gateway Timeout
Date: 2024-01-23
Duration: 116 Minutes
Severity: Sev-1

=====

1. Problem Statement

The monitoring system alerted high latency and error rates in the Payment Gateway Timeout module.

Users experienced 503 errors.

2. Root Cause Analysis (5 Whys)

- Why failed? The connection pool was exhausted.
- Why exhausted? Threads were waiting on I/O.
- Why waiting? The external vendor API was slow.
- Why slow? Vendor was undergoing maintenance.
- Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

4. Prevention / Action Items

- [] Implement Hystrix Circuit Breaker (Due: Jan 30)
- [] Increase timeout thresholds (Due: Jan 25)
- [] Add alerting for connection pool saturation.

5. Timeline

08:00 - Alert fired
08:05 - Sev-1 declared
08:15 - Root cause identified
08:30 - Fix deployed

=====

CONFIDENTIAL - INTERNAL USE ONLY

```
=====
Incident Report: INC-403-17
Subject:         DDoS Attack on API
Date:           2024-01-13
Duration:       56 Minutes
Severity:       Sev-1
=====
```

1. Problem Statement

The monitoring system alerted high latency and error rates in the DDoS Attack on API module.

Users experienced 503 errors.

2. Root Cause Analysis (5 Whys)

- Why failed? The connection pool was exhausted.
- Why exhausted? Threads were waiting on I/O.
- Why waiting? The external vendor API was slow.
- Why slow? Vendor was undergoing maintenance.
- Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

4. Prevention / Action Items

- [] Implement Hystrix Circuit Breaker (Due: Jan 30)
- [] Increase timeout thresholds (Due: Jan 25)
- [] Add alerting for connection pool saturation.

5. Timeline

08:00 - Alert fired
08:05 - Sev-1 declared
08:15 - Root cause identified
08:30 - Fix deployed

```
=====
Incident Report: INC-405-18
Subject:         Redis Cache Fragmentation
Date:           2024-01-16
Duration:       71 Minutes
Severity:       Sev-1
=====
```

1. Problem Statement

The monitoring system alerted high latency and error rates in the Redis Cache Fragmentation module.

Users experienced 503 errors.

CONFIDENTIAL - INTERNAL USE ONLY

2. Root Cause Analysis (5 Whys)

- Why failed? The connection pool was exhausted.
- Why exhausted? Threads were waiting on I/O.
- Why waiting? The external vendor API was slow.
- Why slow? Vendor was undergoing maintenance.

- Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

4. Prevention / Action Items

- [] Implement Hystrix Circuit Breaker (Due: Jan 30)
- [] Increase timeout thresholds (Due: Jan 25)
- [] Add alerting for connection pool saturation.

5. Timeline

08:00 - Alert fired

08:05 - Sev-1 declared

08:15 - Root cause identified

08:30 - Fix deployed

=====

=====

Incident Report: INC-401-19

Subject: Database CPU Spike

Date: 2024-01-15

Duration: 12 Minutes

Severity: Sev-1

=====

1. Problem Statement

The monitoring system alerted high latency and error rates in the Database CPU Spike module.

Users experienced 503 errors.

2. Root Cause Analysis (5 Whys)

- Why failed? The connection pool was exhausted.
- Why exhausted? Threads were waiting on I/O.
- Why waiting? The external vendor API was slow.
- Why slow? Vendor was undergoing maintenance.

- Root Cause: Lack of circuit breaker configuration for vendor maintenance windows.

3. Resolution

Restarted the pods and enabled the fallback mocked response feature.

CONFIDENTIAL - INTERNAL USE ONLY

4. Prevention / Action Items

- [] Implement Hystrix Circuit Breaker (Due: Jan 30)
- [] Increase timeout thresholds (Due: Jan 25)
- [] Add alerting for connection pool saturation.

5. Timeline

08:00 - Alert fired

08:05 - Sev-1 declared

08:15 - Root cause identified

08:30 - Fix deployed

=====